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Solving the Time-Dependent Schrödinger Equation

via the Crank–Nicolson Method: Stability, Unitarity & Efficient Linear Algebra

A complete mathematical derivation from operator formalism to tridiagonal linear systems and the Thomas algorithm

GitHub

1. The Physical Problem

The **Time-Dependent Schrödinger Equation** (TDSE) governs quantum state evolution:

$$\boxed{i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \hat{H}\,\Psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\Psi(x,t)} \tag{1}$$

Formal solution - time-evolution operator:

$$\Psi(x, t+\Delta t) = \underbrace{e^{-i\hat{H}\Delta t/\hbar}}_{\hat{U}(\Delta t)}\,\Psi(x,t)$$

Unitarity: the Core Requirement

$\hat{H} = \hat{H}^\dagger$ (Hermitian) $\Rightarrow \hat{U}^\dagger\hat{U} = e^{+i\hat{H}\Delta t/\hbar}e^{-i\hat{H}\Delta t/\hbar} = \hat{I}$

This enforces **probability conservation**:

$$\int_{-\infty}^{+\infty}|\Psi(x,t)|^2\,dx = 1 \quad \forall t$$

Any valid numerical scheme must discretely replicate this.

The computational challenge: approximate $e^{-i\hat{H}\Delta t/\hbar}$ on a finite spatial grid while keeping the discrete propagator *exactly* unitary.

2. Space–Time Discretization

Introduce a uniform grid with $N+1$ spatial and M temporal points:

$$x_j = j\,\Delta x,\quad j = 0,\dots,N; \qquad t_n = n\,\Delta t,\quad n = 0,1,\dots$$

Central finite difference (2^{nd}-order, $\mathcal{O}(\Delta x^2)$):

$$\left.\frac{\partial^2\Psi}{\partial x^2}\right|_{x_j} \approx \frac{\Psi_{j+1}^n - 2\Psi_j^n + \Psi_{j-1}^n}{(\Delta x)^2} \tag{2}$$

Derived by Taylor expansion: $\Psi_{j\pm1} = \Psi_j \pm \Delta x\Psi' + \frac{(\Delta x)^2}{2}\Psi'' \pm \frac{(\Delta x)^3}{6}\Psi''' + \mathcal{O}(\Delta x^4)$

The **discrete Hamiltonian** H_{FD} acts on the grid vector $\boldsymbol{\Psi}^n = [\Psi_1^n,\dots,\Psi_{N-1}^n]^\mathsf{T}$:

$$(H_{FD}\boldsymbol{\Psi}^n)_j = -\beta\Psi_{j-1}^n + (2\beta + V_j)\Psi_j^n - \beta\Psi_{j+1}^n, \qquad \beta \equiv \frac{\hbar^2}{2m(\Delta x)^2}$$

Boundary Conditions

- **Dirichlet** (infinite well): $\Psi_0^n = \Psi_N^n = 0$
- **Periodic** (ring/lattice): $\Psi_0^n = \Psi_N^n,\ \Psi_{-1}^n = \Psi_{N-1}^n$
- **Absorbing** (open boundary): Add $-i\Gamma(x)$ to $V(x)$ near edges

3. Why Euler Schemes Fail

Stability test: substitute $\Psi_j^n = g^n e^{ikj\Delta x}$ (Fourier mode) into each scheme and compute amplification factor g.

Forward Euler (FTCS)
$\Psi^{n+1} = (I - \frac{i\Delta t}{\hbar}H)\Psi^n$
$|g| = \sqrt{1 + (\frac{\Delta t\lambda}{\hbar})^2} > 1$
Unconditionally unstable.
Norm grows $\to$ blows up.

$\rightarrow$ **Backward Euler (BTCS)**
$\Psi^{n+1} = (I + \frac{i\Delta t}{\hbar}H)^{-1}\Psi^n$
$|g| = \frac{1}{\sqrt{1+(\Delta t\lambda/\hbar)^2}} < 1$
Stable, but **artificially dissipative.**
Norm decays $\Rightarrow$ violates QM.

$\rightarrow$ **Crank–Nicolson**
Average of FTCS & BTCS
$|g| = \left|\frac{1 - i\Delta t\lambda/(2\hbar)}{1 + i\Delta t\lambda/(2\hbar)}\right| = 1$
Unconditionally stable. Exact norm conservation.

Here λ denotes any eigenvalue of H_{FD}; since H_{FD} is real-symmetric all eigenvalues are real, so $|g| = 1$ holds for *every* mode simultaneously - unconditional stability.

4. Crank-Nicolson & the Padé Approximants

Instead of evaluating the derivative at the current *or* future time alone, Crank-Nicolson evaluates the TDSE at the **temporal midpoint** $t_{n+1/2}$ by averaging the Hamiltonian at both levels:

$$i\hbar\frac{\Psi_j^{n+1} - \Psi_j^n}{\Delta t} = \tfrac{1}{2}\left(H_{FD}\Psi_j^{n+1} + H_{FD}\Psi_j^n\right) \tag{3}$$

The exact time-evolution operator is $\hat{U} = e^{-ix}$, where $x \equiv \hat{H}\Delta t/\hbar$. Direct computation of the matrix exponential $e^{-i\hat{H}\Delta t/\hbar}$ is impractically expensive. Instead, the Crank–Nicolson method employs the **Padé [1,1] approximant**:

$$e^{-ix} \approx R(x) = \frac{1 - ix/2}{1 + ix/2} \implies U_{CN} = \left(I + i\frac{\Delta t}{2\hbar}H\right)^{-1}\left(I - i\frac{\Delta t}{2\hbar}H\right) \tag{4}$$

Taylor Series Matching (Order of Accuracy)

Let's compare their expansions for a small time step x:

$$\text{Exact:}\quad e^{-ix} = 1 - ix - \frac{x^2}{2} + i\frac{x^3}{6} + \mathcal{O}(x^4)$$

$$\text{Padé:}\quad R(x) = \left(1 - i\frac{x}{2}\right)\left(1 - i\frac{x}{2} - \frac{x^2}{4} + i\frac{x^3}{8} + \dots\right)$$

$$= 1 - ix - \frac{x^2}{2} + i\frac{x^3}{4} + \mathcal{O}(x^4)$$

The expressions **match exactly up to** $\mathcal{O}(x^2)$. The discrepancy begins at the $\mathcal{O}(x^3)$ term, meaning the local truncation error per step is $\mathcal{O}(\Delta t^3)$. This makes the method **globally 2nd-order accurate** in time.

5. Tridiagonal Matrix Structure of A and B

Let $r_j \equiv \gamma(2\beta + V_j)$ (diagonal) and $s \equiv \gamma\beta$ (off-diagonal) with $\gamma = \Delta t/(2\hbar)$. The linear system (5) reads:

$$\underbrace{\begin{pmatrix} 1{+}ir_1 & -is & 0 & \cdots & 0 \\ -is & 1{+}ir_2 & -is & \cdots & 0 \\ 0 & -is & 1{+}ir_3 & \cdots & 0 \\ \vdots & & & \ddots & -is \\ 0 & 0 & \cdots & -is & 1{+}ir_{N-1} \end{pmatrix}}_{A}\boldsymbol{\Psi}^{n+1} = \underbrace{\begin{pmatrix} 1{-}ir_1 & +is & 0 & \cdots & 0 \\ +is & 1{-}ir_2 & +is & \cdots & 0 \\ 0 & +is & 1{-}ir_3 & \cdots & 0 \\ \vdots & & & \ddots & +is \\ 0 & 0 & \cdots & +is & 1{-}ir_{N-1} \end{pmatrix}}_{B}\boldsymbol{\Psi}^n \tag{5}$$

Expanding the Discrete Hamiltonian

Substituting $H_{FD}\Psi_j = -\beta\Psi_{j-1} + (2\beta + V_j)\Psi_j - \beta\Psi_{j+1}$ into both sides of (5), grouping by spatial index $j-1,\ j,\ j+1$:

Left side (future unknowns Ψ^{n+1}):

$$(-i\gamma\beta)\,\Psi_{j-1}^{n+1} + [1 + i\gamma(2\beta + V_j)]\,\Psi_j^{n+1} + (-i\gamma\beta)\,\Psi_{j+1}^{n+1}$$

Right side (present knowns Ψ^n):

$$(+i\gamma\beta)\,\Psi_{j-1}^n + [1 - i\gamma(2\beta + V_j)]\,\Psi_j^n + (+i\gamma\beta)\,\Psi_{j+1}^n$$

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6. Thomas Algorithm - $\mathcal{O}(N)$ Tridiagonal Solver

Naïve LU decomposition costs $\mathcal{O}(N^3)$. The Thomas algorithm exploits the **tridiagonal structure** to achieve $\mathcal{O}(N)$ with only $\approx 8N$ floating-point operations - no fill-in.

Let the system be $a_j\Psi_{j-1} + d_j\Psi_j + c_j\Psi_{j+1} = f_j$ with $a_j = c_j = -is$, $d_j = 1 + ir_j$.

Step 1 - Forward Sweep (modify coefficients)

Initialise: $\tilde{d}_1 = d_1,\ \tilde{f}_1 = f_1$.
For $j = 2, 3, \dots, N-1$:

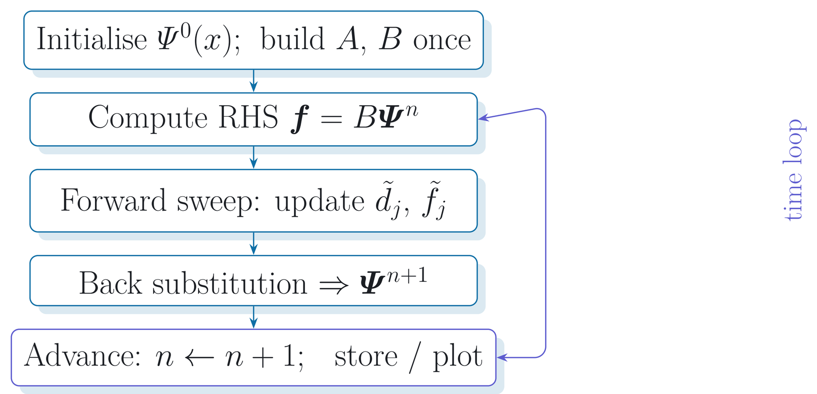
$$m_j = \frac{a_j}{\tilde{d}_{j-1}}, \qquad \tilde{d}_j = d_j - m_j\,c_{j-1}, \qquad \tilde{f}_j = f_j - m_j\,\tilde{f}_{j-1}$$

Eliminates the sub-diagonal in $\mathcal{O}(N)$ steps.

Step 2 - Back Substitution (recover $\boldsymbol{\Psi}^{n+1}$)

$$\Psi_{N-1}^{n+1} = \frac{\tilde{f}_{N-1}}{\tilde{d}_{N-1}}, \qquad \Psi_j^{n+1} = \frac{\tilde{f}_j - c_j\Psi_{j+1}^{n+1}}{\tilde{d}_j}, \qquad j = N-2,\dots,1$$

Full time-stepping algorithm:



7. Method Comparison at a Glance

Scheme	Stability	Norm	Accuracy	Cost/step
Forward Euler (FTCS)	Unstable	Grows	$\mathcal{O}(\Delta t, \Delta x^2)$	$\mathcal{O}(N)$
Backward Euler (BTCS)	Stable	Decays	$\mathcal{O}(\Delta t, \Delta x^2)$	$\mathcal{O}(N)$
Crank–Nicolson	**Stable**	**Exact**	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$\mathcal{O}(N)$
Dense LU solve	Stable	Exact	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$\mathcal{O}(N^3)$

CN with Thomas algorithm delivers the best possible combination: $\mathcal{O}(N)$ cost, $\mathcal{O}(\Delta t^2)$ accuracy, and exact unitarity - simultaneously.

8. Summary & Physical Significance

Crank–Nicolson Recipe

1. **Setup:** set grid Δx, Δt and $\gamma = \Delta t/(2\hbar)$.
2. **Matrices:** build tridiagonal $A, B = I \pm i\gamma H_{FD}$ just *once*.
3. **Iterate:** solve $A\boldsymbol{\Psi}^{n+1} = B\boldsymbol{\Psi}^n$ using the $\mathcal{O}(N)$ Thomas Algorithm.

Why CN is the Gold Standard

- **Unitary:** Padé(1,1) approximant exactly conserves probability ($\|\boldsymbol{\Psi}\| = 1$).
- **Stable:** Unconditionally stable without artificially dissipating the wave-packet.
- **Accurate:** $\mathcal{O}(\Delta t^2, \Delta x^2)$ – second-order accuracy in space and time.